# Year 11 Mathematics

## Algebra 9

7.2 Formulae to Solve Inverse Proportion Problems

# Formulae to Solve Inverse Proportion Problems

STARTER · 5 MIN

**Date:** _____ **Title:** Formulae to Solve Inverse Proportion Problems

**LEARNING OBJECTIVES**

- Recognise inverse proportion and use $y = k \div x$
- Use inverse proportion with squares and square roots
- Solve problems involving inverse proportion

**STARTER** *Answer in your book. No calculators.*

1. Solve $4 = k \div 3$
2. Work out $36 \div 9$
3. 3 workers take 8 days. How many worker-days is this?
4. Work out $5^2$

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. $k = \mathbf{12}$

2. **4**

3. $3 \times 8 = \mathbf{24}$

4. **25**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can recognise inverse proportion and use $y = k \div x$ | ☐ |
| SILVER | I can use inverse proportion with squares and square roots | ☐ |
| GOLD | I can solve problems involving inverse proportion | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Inverse Proportion

INTRODUCTION · 3 MIN

If y is inversely proportional to x, then as x doubles, y halves.

## Inverse

One goes up, the other goes down.

More workers, fewer days

## Formula

$y \propto \frac{1}{x}$ becomes

$$y = \frac{k}{x}$$

## Find k

$k = x \times y$

$y = 4$ when $x = 3 \rightarrow k = 12$

## Other powers

$y \propto \frac{1}{x^2}$ becomes

$$y = \frac{k}{x^2}$$

# Examples 1 and 2 — Finding the Formula

EXAMPLES · 5 MIN

BRONZE

*Watch and copy into your book*

**Hint** $y \propto \frac{1}{x} \rightarrow y = \frac{k}{x}$

## Example 1

y is inversely proportional to x. y = 6 when x = 4. Find a formula for y in terms of x.

## Example 2

y = 24 ÷ x. Find y when x = 3, and x when y = 2.

# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

$$y = \frac{k}{x}$$

$$6 = k \div 4$$

$$k = 24$$

$$y = \mathbf{\frac{24}{x}}$$

Tip: k = x × y.

## Example 2

$$y = 24 \div 3$$

$$= 8$$

$$2 = 24 \div x$$

$$\mathbf{x = 12}$$

Tip: substitute, then solve.

# Task 1

PRACTICE · 9 MIN

BRONZE *y is inversely proportional to x*

**Hint** $y \propto \frac{1}{x} \rightarrow y = \frac{k}{x}$

1. y = 5 when x = 4. Find k.
2. y = 3 when x = 10. Find a formula.
3. Use y = 36 ÷ x to find y when x = 9.
4. y = 8 when x = 2. Find y when x = 4.
5. 4 workers build a wall in 6 days. How long do 8 workers take?

★ y = 2.5 when x = 6. Find x when y = 5.

# Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1 $k = 5 \times 4 = \mathbf{20}$

2 $k = 30, \quad y = \dfrac{\mathbf{30}}{\mathbf{x}}$

3 $y = 36 \div 9 = \mathbf{4}$

4 $k = 16, \quad y = 16 \div 4 = \mathbf{4}$

5 $4 \times 6 = 24, \quad 24 \div 8 = \mathbf{3\ days}$

★ $k = 15, \quad x = 15 \div 5 = \mathbf{3}$

# Examples 3 and 4 — Squares and Square Roots

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

**Hint** $y \propto \frac{1}{x^2} \rightarrow y = \frac{k}{x^2}$

## Example 3

$y \propto \frac{1}{x^2}$. $y = 5$ when $x = 2$.

Find y when x = 4.

## Example 4

$y \propto 1/\sqrt{x}$. $y = 6$ when $x = 4$.

Find y when x = 9.

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

$$y = \frac{k}{x^2}$$

$5 = k \div 4$

$k = 20$

$y = 20 \div 16$

$\mathbf{= 1.25}$

Tip: square x first.

## Example 4

$y = k \div \sqrt{x}$

$6 = k \div 2$

$k = 12$

$y = 12 \div 3$

$\mathbf{= 4}$

Tip: $\sqrt{4} = 2$.

# Task 2

PRACTICE · 9 MIN

SILVER

*Answer in your book*

**Hint** $y \propto \frac{1}{x^2} \rightarrow y = \frac{k}{x^2}$

1. $y \propto \frac{1}{x^2}$. $y = 4$ when $x = 3$. Find $y$ when $x = 6$.

2. $y \propto \frac{1}{x^2}$. $y = 10$ when $x = 2$. Find $y$ when $x = 5$.

3. $y \propto 1/\sqrt{x}$. $y = 3$ when $x = 16$. Find $y$ when $x = 4$.

4. $y \propto \frac{1}{x^2}$. $y = 2$ when $x = 5$. Find $x$ when $y = 8$ $(x > 0)$.

5. $F \propto \frac{1}{d^2}$. $F = 100$ when $d = 1$. Find $F$ when $d = 5$.

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

1. $k = 36, \quad y = 36 \div 36 = \mathbf{1}$

2. $k = 40, \quad y = 40 \div 25 = \mathbf{1.6}$

3. $k = 12, \quad y = 12 \div 2 = \mathbf{6}$

4. $k = 50, \quad x^2 = 6.25, \quad x = \mathbf{2.5}$

5. $k = 100, \quad F = 100 \div 25 = \mathbf{4}$

# Examples 5 and 6 — Inverse Proportion Problems

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

**Hint** $y \propto \frac{1}{x^2} \rightarrow y = \frac{k}{x^2}$

## Example 5

(3 marks)

The force F between two magnets is inversely proportional to the square of the distance d. F = 40 when d = 3. Find F when d = 6.

## Example 6

(2 marks)

$y \propto \frac{1}{x^2}$. What happens to y when x is doubled?

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

$$F = \frac{k}{d^2}$$

$$40 = k \div 9$$

$$k = 360$$

$$F = 360 \div 36$$

$$\mathbf{= 10}$$

Tip: write the formula first.

## Example 6

Replace x with 2x

$$y = \frac{k}{4x^2}$$

**y is divided by 4**

Tip: replace x with 2x.

# Task 3

PRACTICE · 9 MIN

GOLD

*Exam-style: show all your working*

**Hint** $y \propto \frac{1}{x^2} \rightarrow y = \frac{k}{x^2}$

1. The time t to fill a tank is inversely proportional to the number of pipes n. 3 pipes take 40 minutes. How long do 5 pipes take? (3 marks)

2. Light intensity I is inversely proportional to the square of distance d. I = 50 when d = 2. Find I when d = 10. (3 marks)

3. $y \propto \frac{1}{x^2}$. What happens to y when x is halved? (2 marks)

4. $y \propto \frac{1}{x}$. y = 12 when x = 5. Find x when y = 20. (2 marks)

5. $y \propto \frac{1}{\sqrt{x}}$. y = 10 when x = 9. Find x when y = 6. (3 marks)

# Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $k = 120, \quad t = 120 \div 5 =$ **24 minutes**

2. $k = 200, \quad l = 200 \div 100 =$ **2**

3. $(½)^2 = ¼,$ **y is multiplied by 4**

4. $k = 60, \quad x = 60 \div 20 =$ **3**

5. $k = 30, \quad \sqrt{x} = 5, \quad x =$ **25**

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** $y \propto \frac{1}{x}$. $y = 4$ when $x = 5$. Find $k$.

**SILVER** $y = 48 \div x$. Find $y$ when $x = 6$.

**GOLD** $y \propto \frac{1}{x^2}$. $y = 9$ when $x = 2$. Find $y$ when $x = 3$.

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

**BRONZE** $k = 4 \times 5 = \mathbf{20}$

**SILVER** $y = 48 \div 6 = \mathbf{8}$

**GOLD** $k = 36, \quad y = 36 \div 9 = \mathbf{4}$

# Exam Practice — Inverse Proportion

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: show all your working*

**Hint** $y \propto \frac{1}{x} \rightarrow y = \frac{k}{x}$

## Question 1

(3 marks)

y is inversely proportional to x.

y = 8 when x = 3.

**(a)** Find a formula for y in terms of x. (2)

.....

**(b)** Find y when x = 6. (1)

y = .....

## Question 2

(3 marks)

The time, t days, to build a wall is inversely proportional to the number of builders, n.

4 builders take 9 days.

How long do 6 builders take?

..... days

# Exam Practice — Squares and Square Roots

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint** $y \propto \frac{1}{x^2} \rightarrow y = \frac{k}{x^2}$

## Question 3

(3 marks)

y is inversely proportional to $x^2$.

y = 9 when x = 2.

Find y when x = 3.

y = .....

## Question 4

(3 marks)

y is inversely proportional to the square root of x.

y = 10 when x = 4.

Find y when x = 25.

y = .....

# Exam Practice — Inverse Proportion Problems

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** $y \propto \frac{1}{x^2} \rightarrow y = \frac{k}{x^2}$

## Question 5

(4 marks)

The light intensity, I, is inversely proportional to the square of the distance, d.

I = 80 when d = 2.

**(a)** Find I when d = 4. (2)

I = .....

**(b)** Find d when I = 5 (d > 0). (2)

d = .....

## Question 6

(2 marks)

y is inversely proportional to $x^2$.

x is halved.

What happens to y?

.....

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

$y = \frac{k}{x}$, $8 = \frac{k}{3}$ — M1

**$k = 24$, $y = \frac{24}{x}$** — A1

**(b) $y = 24 \div 6 = 4$** — B1

Tip: $k = x \times y$.

## Question 2

$k = 4 \times 9 = 36$ — M1

$t = \frac{36}{n}$ — M1

**$t = 36 \div 6 = 6$ days** — A1

Tip: more builders, fewer days.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

$y = \frac{k}{x^2}$, $9 = \frac{k}{4}$ — M1

$k = 36$ — M1

**$y = 36 \div 9 = 4$** — A1

Tip: square x first.

## Question 4

$y = \frac{k}{\sqrt{x}}$, $10 = \frac{k}{2}$ — M1

$k = 20$ — M1

**$y = 20 \div 5 = 4$** — A1

Tip: the square root of 25 is 5.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

$I = \frac{k}{d^2}$, $k = 80 \times 4 = 320$ — M1

**(a) $I = 320 \div 16 = 20$** — A1

(b) $5 = 320 \div d^2$, $d^2 = 64$ — M1

**$d = 8$** — A1

Tip: write the formula first.

## Question 6

$x \div 2$ makes $x^2$ divide by 4 — M1

**y is multiplied by 4** — A1

Tip: halving x divides $x^2$ by 4.